

Find large files recursively in a directory tree

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Recursively finds all regular files larger than 100 MB in the directory tree, displays them with human-readable sizes, and sorts by size in descending order.

Language

bash

Category

filesystem

Command

```
find . -type f -size +100M -exec ls -lh {} + | awk '{print $5, $9}' | sort -hr
```

Explanation

The command uses 'find' to recursively traverse the directory tree, filter for regular files (-type f), and select those exceeding 100 MB (-size +100M). The '-exec ls -lh {} +' displays detailed information for each file in a batch operation, which is more efficient than using '\{};'. The output is piped to 'awk' to extract the file size and path, then 'sort -hr' sorts the results by human-readable sizes in descending order, showing the largest files first.

Tags

find, filesystem, disk-usage, large-files, recursive, storage

Dependencies

findutils, coreutils

Arguments

1. **PATH** (Optional): Root directory to start the recursive search (default: current directory)
Default: .

2. **SIZE** (Optional): Minimum file size threshold (default: 100M, supports K, M, G suffixes)
Default: 100M

Examples

1. `find . -type f -size +100M -exec ls -lh {} + | awk '{print $5, $9}' | sort -hr` - Find files larger than 100 MB in current directory and subdirectories
2. `find /var -type f -size +500M -exec ls -lh {} + | awk '{print $5, $9}' | sort -hr` - Find large files in /var directory
3. `find . -type f -size +1G -exec ls -lh {} + | awk '{print $5, $9}' | sort -hr` - Find files larger than 1 GB
4. `find . -type f -size +50M -exec ls -lh {} + | awk '{print $5, $9}' | sort -hr | head -20` - Show top 20 largest files over 50 MB

Output

```
1.5G ./videos/archive.tar.gz
987M ./backups/database.sql.bz2
756M ./datasets/training-data.bin
543M ./cache/node_modules.tar
387M ./downloads/ubuntu-22.04-iso
256M ./logs/app-2026.log
198M ./tmp/large-temp-file
```

Notes

- The -size option supports various units: c (bytes), k (kilobytes), M (megabytes), G (gigabytes), and b (512-byte blocks)
- By default, -size rounds up (e.g., +100M matches files larger than 100 MiB)
- Use -size -100M to find files smaller than 100 MB
- The command may take considerable time on large directory trees
- Using '{ } +' instead of '{ } \{\};' is more efficient as it batches the command executions
- Results may be incomplete in directories where you lack read permissions

Warnings

- On network filesystems (NFS, SMB), this command may be significantly slower
- Does not follow symbolic links by default; add -L flag if needed
- Very large directory trees may cause the command to run for extended periods
- Some filesystems may not report accurate sizes for sparse files

See Also

- `find-largest-storage-users`
- `disk-space-sort-largest-directories`

Status

reviewed

Safety

safe

Shell

bash

Platforms

linux, gnu-linux, freebsd, openbsd, netbsd

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